



Girraween High School

2024 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General Instructions

- Reading time: 10 minutes
- Working time: 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- Answer multiple-choice questions on the multiple-choice answer sheet provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks: 100

Section I 10 Marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II 90 marks

- Attempt Questions 11 - 34
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

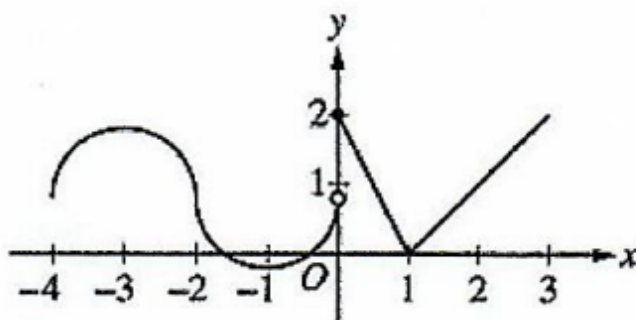
Question 1

What is the gradient of the tangent to the curve $y = \sqrt{10 - x}$ at $x = 1$?

- (A) $\frac{1}{6}$
- (B) $-\frac{1}{6}$
- (C) 6
- (D) -6

Question 2

The graph of the piecemeal function $y = f(x)$ is shown below. The part of the graph in the domain $-4 \leq x < 0$ comprises two semi-circles, joined at $x = -2$. What are the values x for $-4 < x < 3$, at which the function is continuous but not differentiable?



- (A) $x = 1$ only
- (B) $x = -2$ and $x = 0$
- (C) $x = -2$ and $x = 1$
- (D) $x = 0$ and $x = 1$

Examination continues next page...

Question 3

What is the solution of $3^{2x} = 5$?

(A) $x = \frac{2 \ln 5}{\ln 3}$

(B) $x = \frac{\ln 5}{2 \ln 3}$

(C) $x = \frac{2 \ln 3}{\ln 5}$

(D) $x = \frac{\ln 3}{2 \ln 5}$

Question 4

If $a > 0$ and the function $f(x) = ax^3 + bx^2 + cx + d$ is always increasing, what is the condition for a , b and c ?

(A) $b^2 - 2ac > 0$

(B) $b^2 - 2ac < 0$

(C) $b^2 - 3ac > 0$

(D) $b^2 - 3ac < 0$

Question 5

What is the number of solution(s) of the equation $\log_e |x^2 - 1| = 0$?

(A) 2

(B) 3

(C) 1

(D) 0

Examination continues next page...

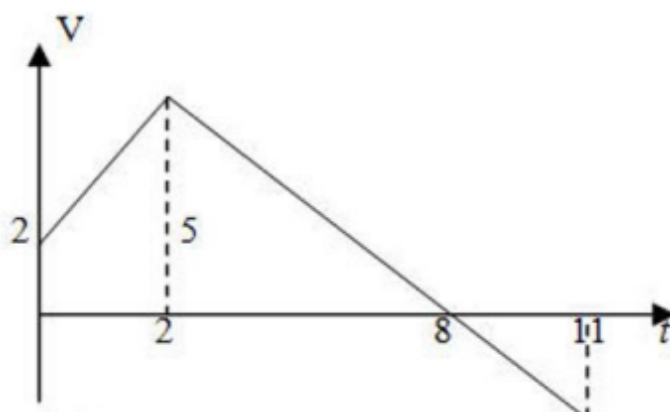
Question 6

Given that $f(x) = \frac{4x^5 - 8x}{x^3}$, what is the value of $f'(2)$?

- (A) 2
- (B) 8
- (C) 12
- (D) 18

Question 7

The diagram gives the velocity of a particle moving in a straight line for 11 seconds.



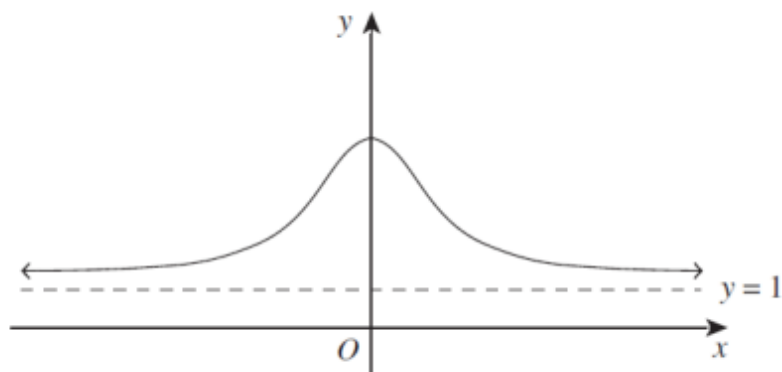
At $t = 11$, the particle is

- (A) 7 metres to the right of the starting point.
- (B) 15 metres to the right of the starting point.
- (C) 22 metres to the right of the starting point.
- (D) 18.25 metres to the right of the starting point.

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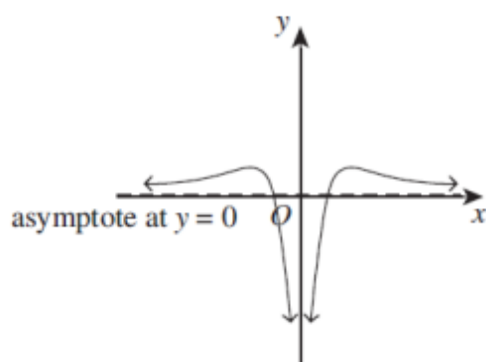
Question 8

The graph shows the function $y = f(x)$ with a horizontal asymptote at $y = 1$.

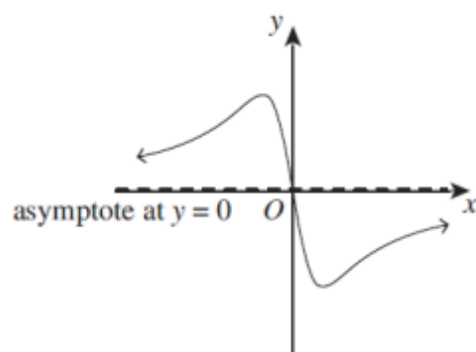


Which of the following could be the graph of $y = f'(x)$?

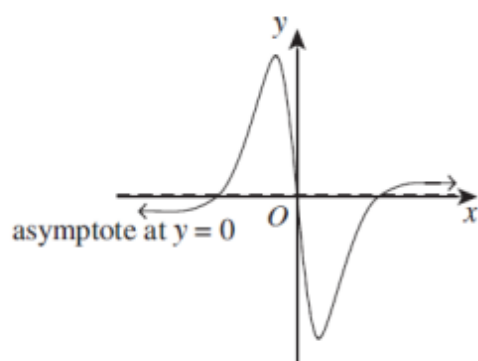
(A)



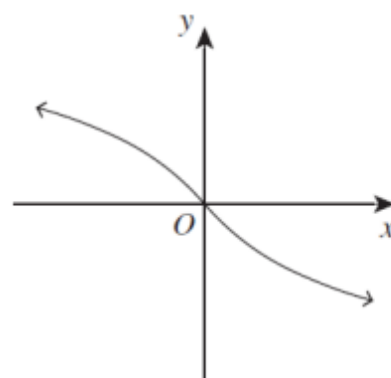
(B)



(C)



(D)



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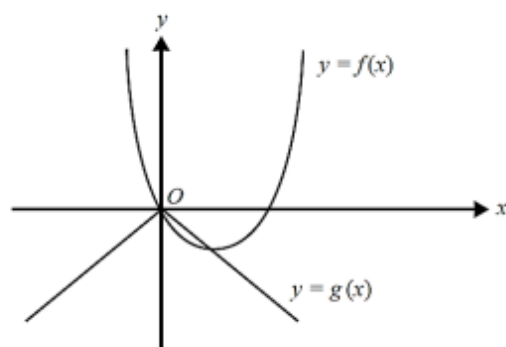
Question 9

Given $\int_2^6 f(x) \, dx = 3$, what is the value of $\int_4^6 f(2(x-3)) \, dx$?

- (A) $\frac{1}{2}$
- (B) $\frac{3}{2}$
- (C) 3
- (D) 8

Question 10

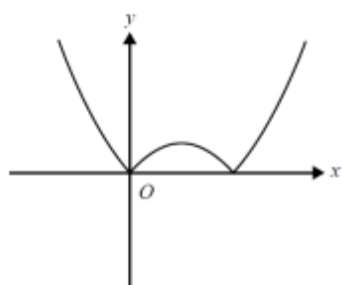
The graph of $y = f(x)$ and $y = g(x)$ are shown below.



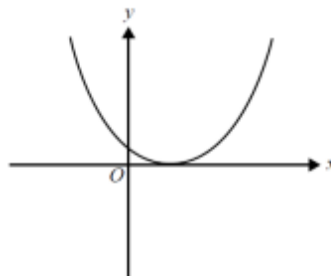
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The graph of $y = f(g(x))$ is best represented by

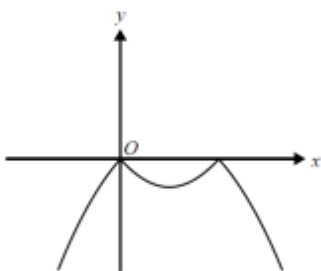
(A)



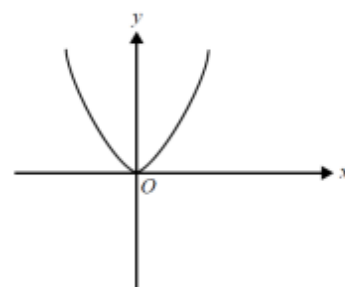
(B)



(C)



(D)



Examination continues next page...

Section II

90 marks

Attempt Questions 11–34

Allow about 2 hours and 45 minutes for this section

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the end of this booklet. If you use this space, clearly indicate which question you are answering.

Question 11

Solve $|4 - 2x| = 3$

2

Question 12

Sketch $y = \frac{3}{x+2} - 1$, showing all intercepts with the coordinate axes and all asymptotes.

3

Examination continues next page...

Question 13Differentiate with respect to x :

a) $y = (3x^2 + 7)^6$ 2

b) $y = 3x^2 \sin x$ 2

Question 14

a) $\int \frac{e^{2x+1}}{e^{2x}} dx$ 2

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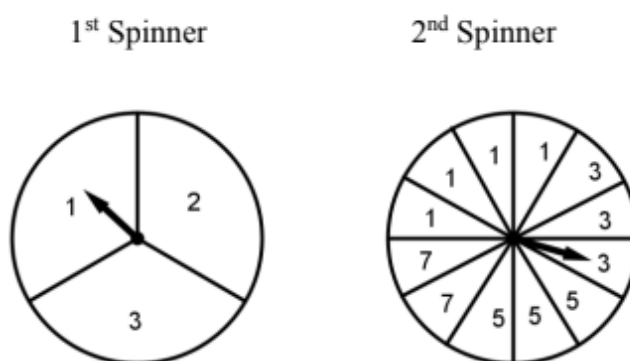
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Question 15

The diagram below shows two spinners.

2



Each of the three outcomes on the first spinner are equally likely.

On the second spinner there are 12 equally likely sectors for the arrow to land on with four possible outcomes.

In a game, both spinners are spun simultaneously.

The players score is the sum of the two numbers that the spinners land on.

In the diagram above, the player's score would be 4 since the first spinner landed on 1 and the second spinner landed on 3.

A player wins if their score is an odd number greater than 6.

What is the probability that a player will win on a single turn?

Examination continues next page...

Question 16

Jim tosses 2 dice with faces numbered 1 to 6. He records the maximum of the two upper faces as a score. The sample space is shown below.

Dice 1 \ Dice 2	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	2	3	4	5	6
3	3	3	3	4	5	6
4	4	4	4	4	5	6
5	5	5	5	5	5	6
6	6	6	6	6	6	6

- a) Jim plays a game with the 2 dice using the results stated above.

2

He wins \$1 if the score is 3 or 4, \$2 if the score is 6 and \$3 if it is 5.

But he loses \$1 if it is 1 or 2.

Complete the probability distribution table for random variable Y showing the probability of winning and losing on each amount.

y	-\$1	\$1	\$2	\$3
$P(Y = y)$				

- b) Find the expected value of Y .

2

Examination continues next page...

- c) If it costs Jim \$2 to play the game, would you expect him to win or lose money? 1

Justify your answer with calculations.

Question 17

For a particular curve it is given that $\frac{d^2y}{dx^2} = 12x - 2$. 3

The tangent at $(1, -2)$ on the curve makes an angle at 45° with the positive direction of the x axis.

Find the equation of the curve.

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Question 18

If $f(x) = \sqrt{x}$ and $g(x) = 4 - x^2$, what is the domain of the function $f(g(x))$?

2

Question 19

The point A (1, 4) lies on the graph $y = f(x)$. When the function is transformed into the graph $y = -2f(x + 1)$, what are the co-ordinates of the point corresponding to point A?

2

Examination continues next page...

Question 20

The energy released by an earthquake, E , can be given by $\log_{10} E = 11.8 + 1.5M$ where M is the measurement of its magnitude on the Richter scale.

- a) Calculate the energy released by an earthquake measuring 6 on the Richter scale. 2

Leave your answer in index form.

- b) If there is another earthquake measuring 3.1 on the Richter scale, how many times more energy is released by the first earthquake than by the second? 2

Answer to the nearest whole number.

Examination continues next page...

Question 21

The closure of several manufacturing businesses in a small town has caused the town's population, P , to decline over the last decade. At the start of 2012 the population was 15 000. At the start of 2022 it was estimated to be 9 500.

Assume that the population decline is proportional to P , so that $\frac{dP}{dt} = -kP$,
where k is a positive constant and t is the time in years.

- a) Show that $P = 15000e^{-kt}$ satisfies the differential equation. 1

- b) Find the value of k in exact form. 2

Examination continues next page...

- c) If the population continues to decrease at the same rate, in what calendar year will the population drop to 5 000?

2

[illegible]

Examination continues next page...

Question 22

A particle starts to move from the origin with a velocity 2 m/s along the x -axis.

Its acceleration at any time t seconds is given by $a = \frac{6}{3t+4}$.

- a) Show that the velocity, v , of the particle at time t is

2

$$v = 2 \ln \left(\frac{3}{4}t + 1 \right) + 2 \text{ m/s}$$

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b) Find the time taken by the particle to reach a velocity $4m/s$.

2

c) Does the particle ever return to the origin? Justify your answer.

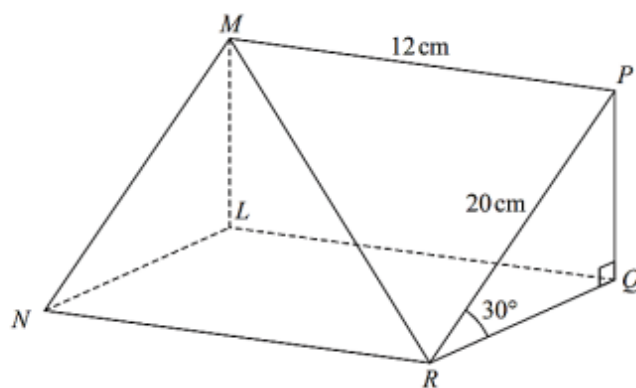
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Examination continues next page...

Question 23

The diagram shows a triangular prism.

3



Not to scale

Triangle PQR is a cross section of the prism.

$PR = 20\text{cm}$, $MP = 12\text{cm}$, $\angle PRQ = 30^\circ$, $\angle PQR = 90^\circ$

Calculate the size of the angle that the line MR makes with the plane $RQLN$. Give your answer correct to 1 decimal place.

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Question 24

The third term of a geometric series is 48 and the sixth term is $\frac{81}{4}$.

- a) Find the common ratio and the first term.

2

- b) Find the sum of the first ten terms to 4 significant figures.

2

Examination continues next page...

Question 25

Consider the function $y = (\ln x)^2$.

- a) Find its stationary point and determine its nature.

3

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- b) Show that there is a point of inflexion at $(e, 1)$.

2

[illegible]

- c) Sketch the function showing all essential features.

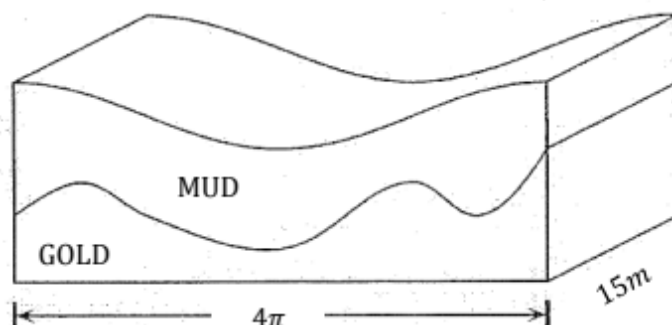
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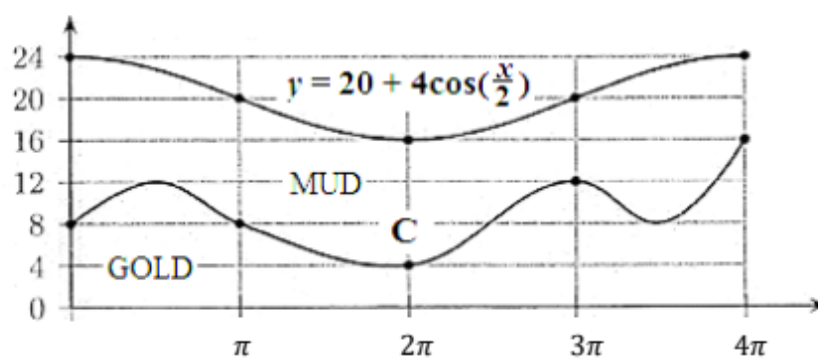
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Question 26

The diagram below shows an amount of gold ore, which is in the shape of a prism underneath a large amount of mud. The width of the prism is 4π metres and its length is 15 metres.



The graph below shows the cross-section of the prism. The top of the mud is given by the function $y = 20 + 4\cos\frac{x}{2}$ and the top of the gold is shown by the curve C .



- a) Find, by integration, the total area of the cross-section of the prism, ie: the area of both the mud and gold.

2

Examination continues next page...

- b) Using the Trapezoidal Rule with the five function values on the graph, find an estimate for the area of the cross-section of the **gold**. 2

- c) Find the volume of the mud. 1

Examination continues next page...

Question 27

The sum S_n of the first n terms of a certain series is $2n + 3n^2$, for $n \geq 1$.

3

Find an expression for the n th term, T_n of the series.

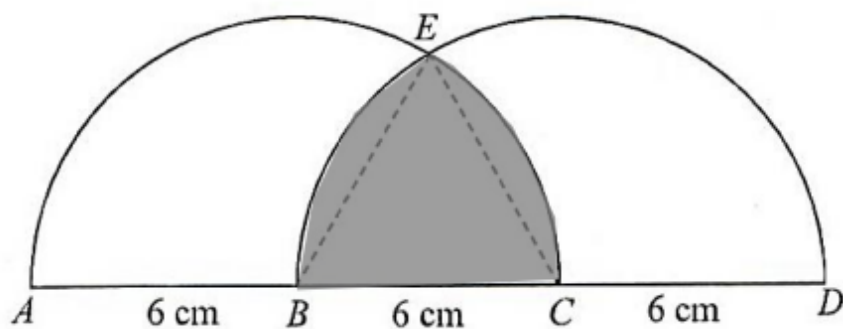
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In the diagram, $ABCD$ is a straight line where $AB = BC = CD = 6\text{cm}$.

The semi-circles on diameter AC and BD intersect at E . $\triangle EBC$ is equilateral.

Find in simplest exact form the area of the shaded region common to the two semi-circles.



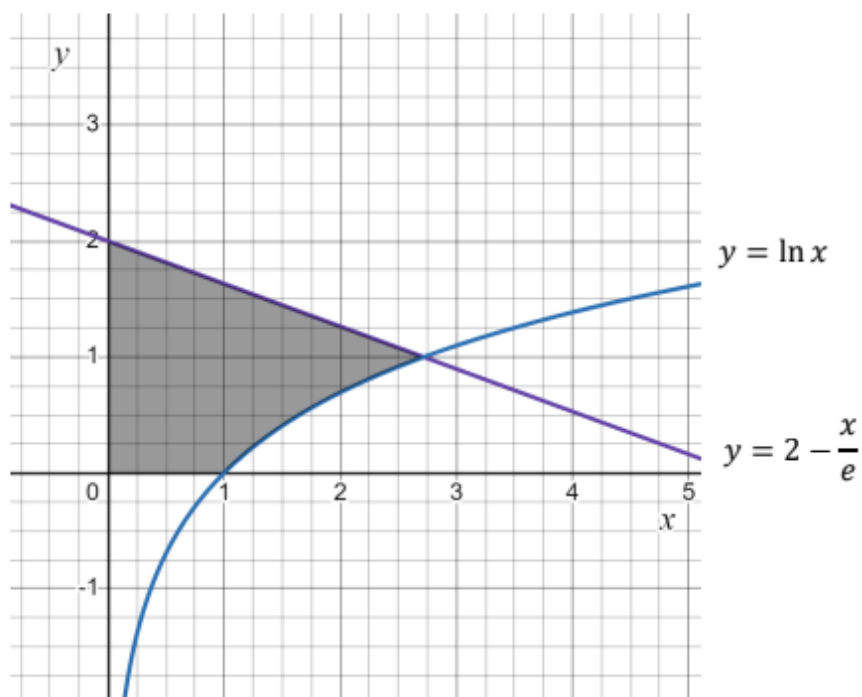
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Examination continues next page...

Question 29

The shaded region below represents the area bounded by the x and y axes, the line

$y = 2 - \frac{x}{e}$ and the curve $y = \ln x$.



- a) Show, by substitution, that the line $y = 2 - \frac{x}{e}$ and the curve $y = \ln x$ intersect at the point $(e, 1)$.

1

Examination continues next page...

b) Hence, find the exact area of the shaded region.

3

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Examination continues next page...

Question 30

- a) Use the change of base law to show the following is a geometric series and find the common ratio.

2

$$\log_3 x + \log_9 x + \log_{81} x + \dots$$

[illegible]

- b) Hence, show that the sum to infinity of the series

2

$$\log_3 x + \log_9 x + \log_{81} x + \dots \text{ is } \log_3 x^2.$$

[illegible]

Examination continues next page...

Question 31

If $x = \frac{e^y - e^{-y}}{2}$ use the substitution $m = e^y$ to solve the equation for y in terms of x .

4

[illegible]

Examination continues next page...

Question 32

A teacher surveyed his class of 24 students about the subjects that they have chosen to study in the following year.

The results indicate that:

- 8 students chose to study neither Physics nor Chemistry
- 8 students chose to study Chemistry
- 12 students chose to study Physics.

- a) Draw a Venn diagram showing this information.

1

- b) Let a student choosing to study Chemistry be event A. Let a student choosing to study Physics be event B. If a student is chosen at random, determine whether events A and B are independent of each other.

2

[illegible]

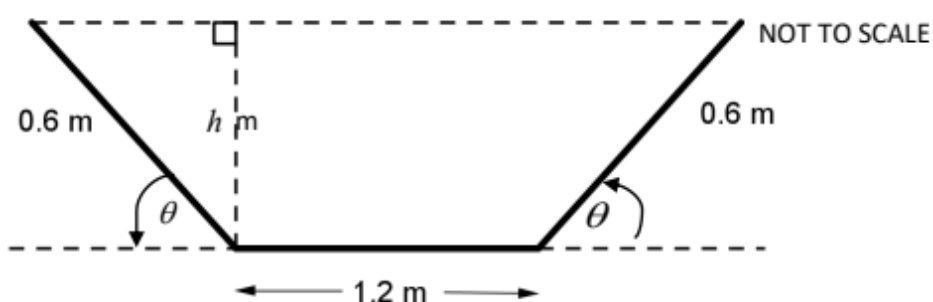
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Question 34

A water overflow channel is to be dug next to a dam. The base of the channel is to be 1.2 metres wide and 15 metres long.

The channel is dug such that the left and right sides are on an angle of θ with the horizontal. The cross-sectional view of the channel is a trapezium with two sides of length 0.6 m.

The diagram below shows a cross-sectional view of the channel.



- a) Find an expression for h in terms of θ .

1

Examination continues next page...

2

[illegible]

33

- c) Find the value of the angle θ so that the volume of the channel is a **maximum**.

Give your answer correct to the nearest minute.

[illegible]

More working space on the next page...

[illegible]

End of examination.